

ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

ADVANCED OBJECT ORIENTED PROGRAMMING (CCS 2122) FINAL EXAM DECEMBER 2024

INSTRUCTIONS

1. This paper consist of seven (7) practical questions and required to answer any five (5) questions.
2. Each question carry equal marks (25)
3. You are required to ensure that you have a working machine and all the programmes are running perfectly.
4. You are required to provide a clear and concise written code as marks will be deducted for poor work.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO

Question One

a) Define a Person class with common attributes, then extend it to create Student and Professor classes with additional unique attributes. **[5 Marks]**

b) Write a program that stores objects of Student and Professor in an array of Person type and uses type casting to access their specific attributes. **[5 Marks]**

c) Discuss the importance of type casting in inheritance when dealing with base class objects. **[5 Marks]**

d) A university management system has different types of users, including Student and Professor. These classes extend from a common base class Person. The system needs to handle these users generically but perform specific operations based on the user type. **[5 Marks]**

Question Two

a) A gaming company in Zambia is designing different types of characters for a role-playing game. The base class Character is abstract, with an abstract method attack() that each character type will implement differently. **[5 Marks]**

b) Define the Character abstract class and include an abstract method attack(). **[5 Marks]**

c) Create two derived classes, Warrior and Mage, that implement the attack() method with unique functionality. **[5 Marks]**

d) Explain the purpose of abstract classes and methods in Java and why they are useful in this context. **[5 Marks]**

Question Three

a) An organization wants to make sure that some critical base classes cannot be extended further. The class Critical System is marked with final to prevent modification. **[5 Marks]**

b) Describe the use of the final keyword in Java and how it applies to inheritance. **[5 Marks]**

c) Explain the implications of marking a class or a method as final. **[5 Marks]**

d) In what situations would marking a class or method as final be beneficial? Provide examples. **[5 Marks]**

Question Four

a) A company has various types of bank accounts, each with a unique interest calculation method. The base class *BankAccount* has a method *calculateInterest()*. Different types of accounts like *SavingsAccount* and *CheckingAccount* override this method to provide specific implementations. **[8 Marks]**

b) Explain polymorphism and how method overriding enables it in Java. **[2 Marks]**

c) Create the **BankAccount** class and derive two classes, **SavingsAccount** and **CheckingAccount**, that override **calculateInterest()**. **[5 Marks]**

d) Write a small program that demonstrates polymorphism by creating an array of **BankAccount** objects and calculating interest for each account type. **[5 Marks]**

Question Five

a) In a vehicle management system, the base class Vehicle has attributes like make, model, and year. The derived class Car has an additional attribute numDoors. **[5 Marks]**

b) Implement the Vehicle and Car classes with constructors for each. Use the super keyword to initialize attributes from the superclass in the Car class. **[5 Marks]**

c) Explain the purpose and use of the super keyword in inheritance with reference to this example. **[5 Marks]**

Question Six

a) Consider a library system where different users have different access levels. A base class User has some attributes marked as private, protected, and public. **[5 Marks]**

b) Describe the member access rules in Java with respect to private, protected, and public access modifiers. **[5 Marks]**

c) Explain how these access modifiers affect inheritance in Java with examples. **[5 Marks]**

d) Create a LibraryMember class that inherits from User and demonstrate how member access is affected by each access modifier. **[5 Marks]**

Question Seven

a) A software development company is working on a project where different types of employees need to be managed. The base class Employee contains basic details like name, employeeID, and salary. Different types of employees, like Manager and Developer, inherit from the Employee class. **[7 Marks]**

b) Define the Employee class with appropriate attributes and a method to display employee details. **[3 Marks]**

c) Using single inheritance, create a Manager class that extends Employee and adds an attribute department. Include methods to set and display department. **[5 Marks]**

d) Explain the advantages of using inheritance in this scenario. **[5 Marks]**